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1. Introduction — Purpose of the FDS

> **What this section covers:** What a Flight Director System is, why it was developed, and its relationship with the autopilot.

The **Flight Director System (FDS)** was originally developed as an aid to the pilot during landing. It provides steering and attitude signals on one instrument, reducing workload. As autopilots became more advanced, FDS signals could be coupled to the autopilot for more complex tasks.

With an FDS, information about **attitude, heading and flight path** can be integrated with navigation information to produce:

- Easy-to-interpret visual instructions for the pilot (via flight director bars), and/or
- Direct input to the autopilot, or both simultaneously

> **FDS has 2 Channels:** Roll channel (first channel) and Pitch channel (second channel). This terminology aligns with autopilot channel descriptions.

2. FDS Information Sources

> **What this section covers:** Inputs to the FDS from various aircraft sensors.

Input	Source
Airspeed, altitude, VSI	Pitot-Static system or Air Data Computer (ADC)
VOR / ILS tracking	VHF Nav receiver
Navigation/routing	FMS, INS/IRS
Attitude (older)	Gyro-magnetic compass + vertical gyro system
Attitude (modern)	INS/IRS (replaces vertical gyro)

3. Flight Director System Components

> **What this section covers:** Each component of a typical modern FDS.

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Electronic Attitude Director Indicator (EADI)

A standard artificial horizon providing pitch and roll. The *Director* part comes from its ability to display demand information from the FDS using **Flight Director Command Bars**. The pilot "flies to" either the intersection of the cross-bars or the point between wedge-shaped pointers. Both displays are intuitive and functionally identical.

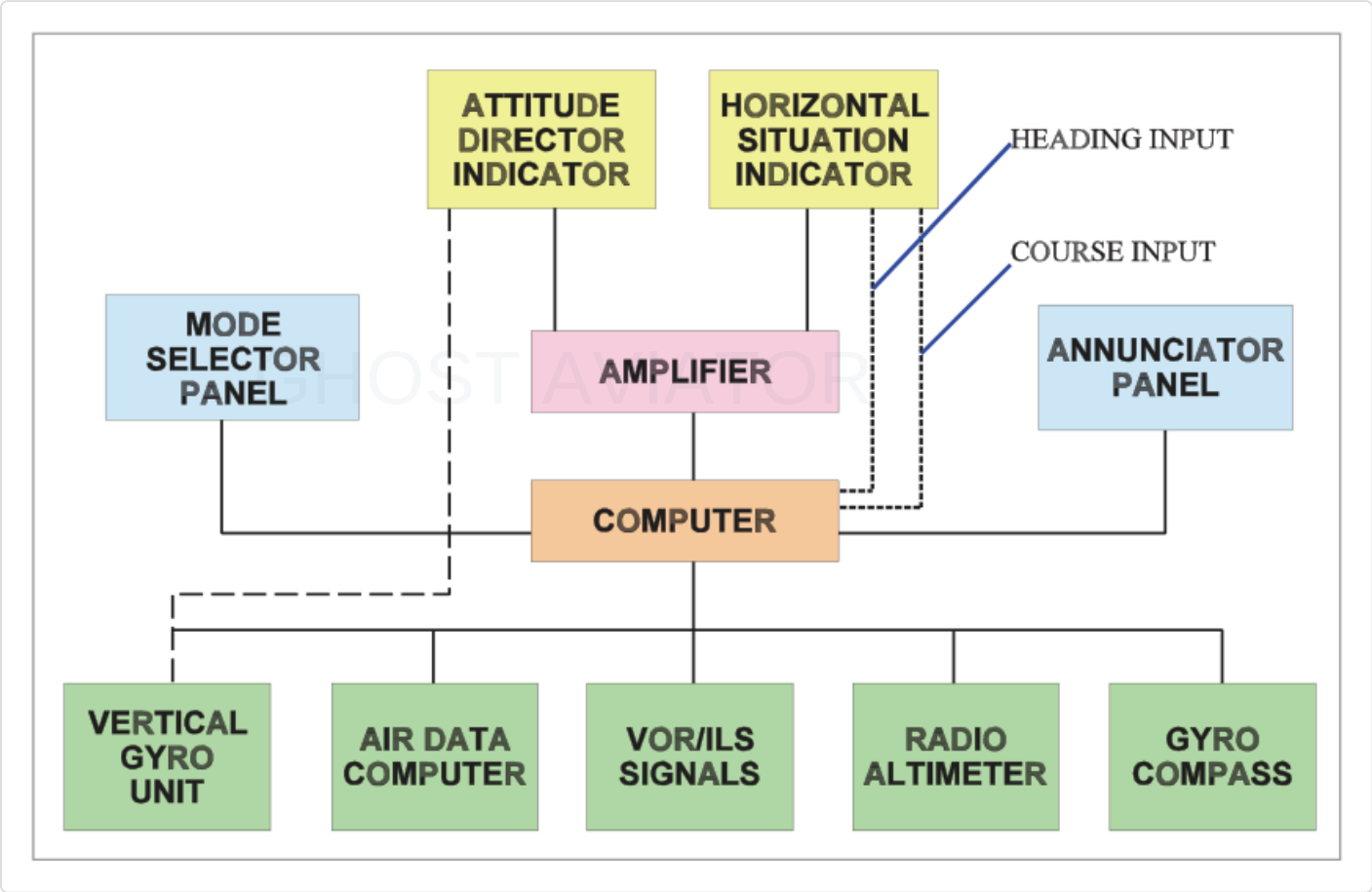


Fig 25.1 — Typical electromechanical FDS / EADI with FD command bars

Primary Flight Display (PFD)

Part of EFIS — brings all flying information onto one display. Contains EADI surrounded by speed, altitude and VSI tapes plus a compass display. Includes the **Flight Mode Annunciator (FMA)** area.

Electronic Horizontal Situation Indicator (EHSI)

A gyro-magnetic compass display with:

- Course Deviation Indicator (CDI) bar
- Deviation dots (scale varies)
- From/to pointer
- Selected course window
- DME display of range
- Heading bug

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Modern aircraft use a **Navigational Display (ND)** and a centralized **Autoflight Mode Control Panel (AMCP / MCP)** for course selection rather than a knob on the HSI.

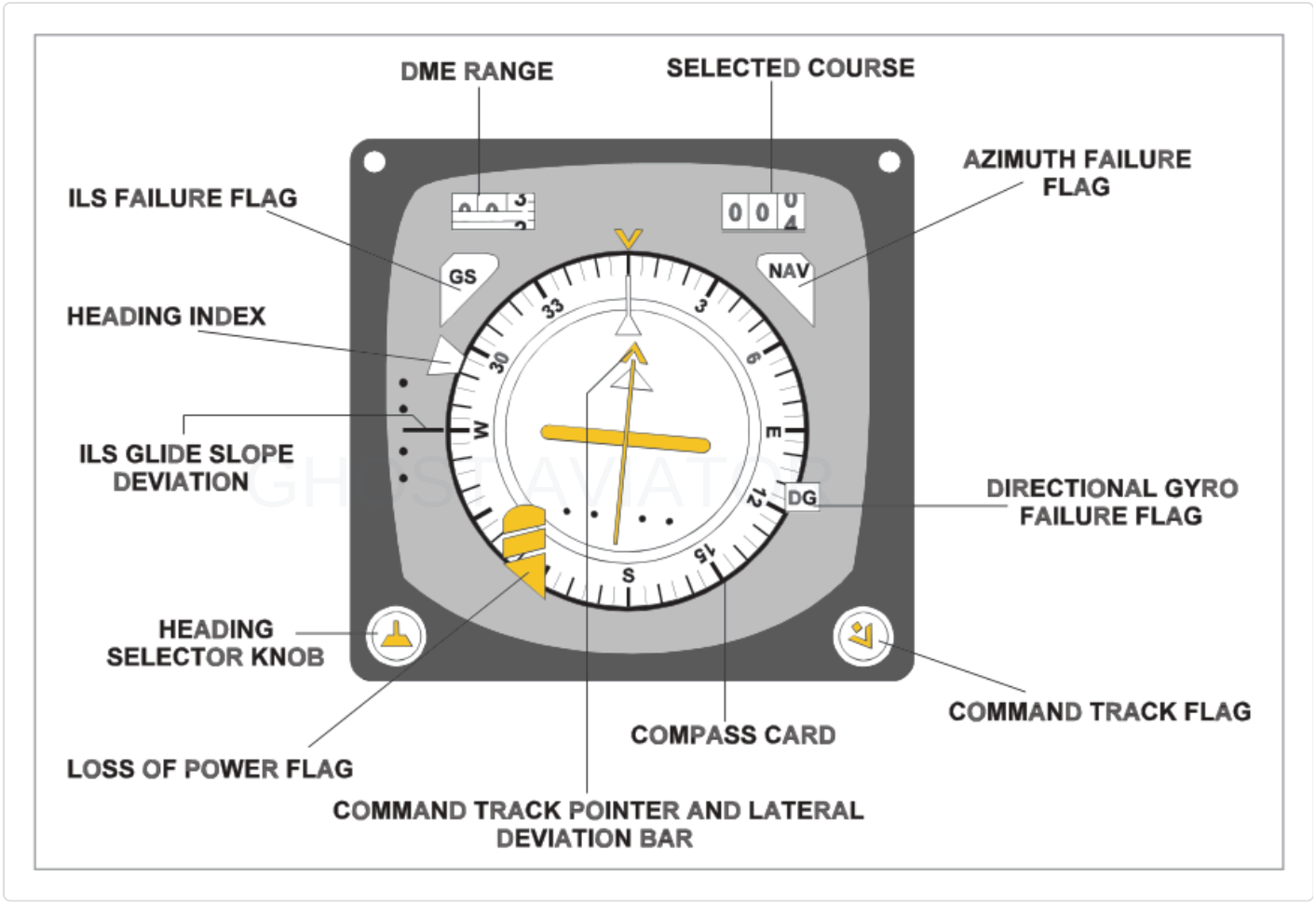


Fig 25.4 — HSI Display

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Fig 25.5 — EHSI Display

Flight Director Computer (FDC)

Gathers and processes all inputs. Older aircraft use analogue ADC/VG outputs; modern systems are purely digital. Outputs go to symbol generators for EADI/EHSI and/or the autopilot.

Optional Components

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Component	Function
Instrument Amplifier / Symbol Generator	Drives electromechanical instrument motors; feeds EFIS symbol generators on modern aircraft
Vertical Gyro (VG)	Remote gyro providing attitude reference on older/smaller aircraft without INS/IRS
INS/IRS	Replaces VG on modern aircraft; more sensitive, near aircraft C of G
Mode Controller / MCP	Allows pilot to change FDS mode, alter pitch trim, switch FDS display on/off
Mode Annunciators / FMA	Shows current FDS/autopilot mode and phase

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FMA (Flight Mode Annunciator) on EFIS Aircraft

- **Engaged modes** — top line, shown in **green**
- **Armed modes** — second line, shown in **white**
- **Newly changed mode** — surrounded by a box for emphasis

On older annunciator panels: LOC light = **amber** (armed) → **green** (captured). GS light = **amber** (armed) → **green** (captured).

4. Raw vs Computed Information

What this section covers: The distinction between raw and computed FDS information — critical for failure scenarios.

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Type	Description	Source
Raw Information	Unprocessed; shows aircraft deviation left/right of track or above/below GS in degrees	ILS glide slope indicator, CDI bar
Computed Information	Processed from rate of change of deviation; anticipates turns/climbs/descents for smooth flying	FDC output to command bars

On FDS Failure: Computed information is no longer reliable, but *raw information may still be available* and can be used for navigation. This is critical in the event of an FDC failure during an ILS approach.

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Exam Tip: Computed information uses the *rate of change* of deviation — as deviation rate increases, the FDS demands a turn to intercept track. Raw information simply shows how far off track you are.

5. FD Fail Indications

What this section covers: How the FDS warns of failures on electromechanical and EFIS displays.

Flag/Warning	Failure Indicated	Location
"GYRO" warning flag	Vertical gyro or vertical reference system failure, or power supply failure	ADI
"ATT" or "FD" flag	FDC failure, instrument amplifier failure, or ADI failure	ADI
"GS" flag	Glide slope information unreliable or system failure	GS scale on ADI or HSI
"NAV" flag	Poor reception, unreliable or loss of VOR, LNAV or LOC information	HSI
Power failure flag	Loss of power to HSI or compass gyro; also indicates DG mode	HSI

DG Mode: The HSI will indicate if the compass system is operating in Directional Gyro (DG) mode — meaning the magnetic reference has been lost and the compass is essentially a DI (Direction Indicator) without magnetic slaving.

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6. Flight Director Modes

What this section covers: The main FDS modes common across aircraft types.

Command Attitude Changes

The FD command bars can be offset vertically to provide an aiming point for maintaining a constant pitch angle (level flight, climb, or descent). Achieved by moving the **Pitch Trim knob/wheel**. Pitch trim is inhibited whenever any other pitch mode is active.

Altitude Acquire / Altitude Hold

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Signals from ADC through FDS maintain selected level. If aircraft deviates above desired altitude, FD commands **pitch down**; if below, it commands **pitch up**. Command bars indicate neutral position at selected level.

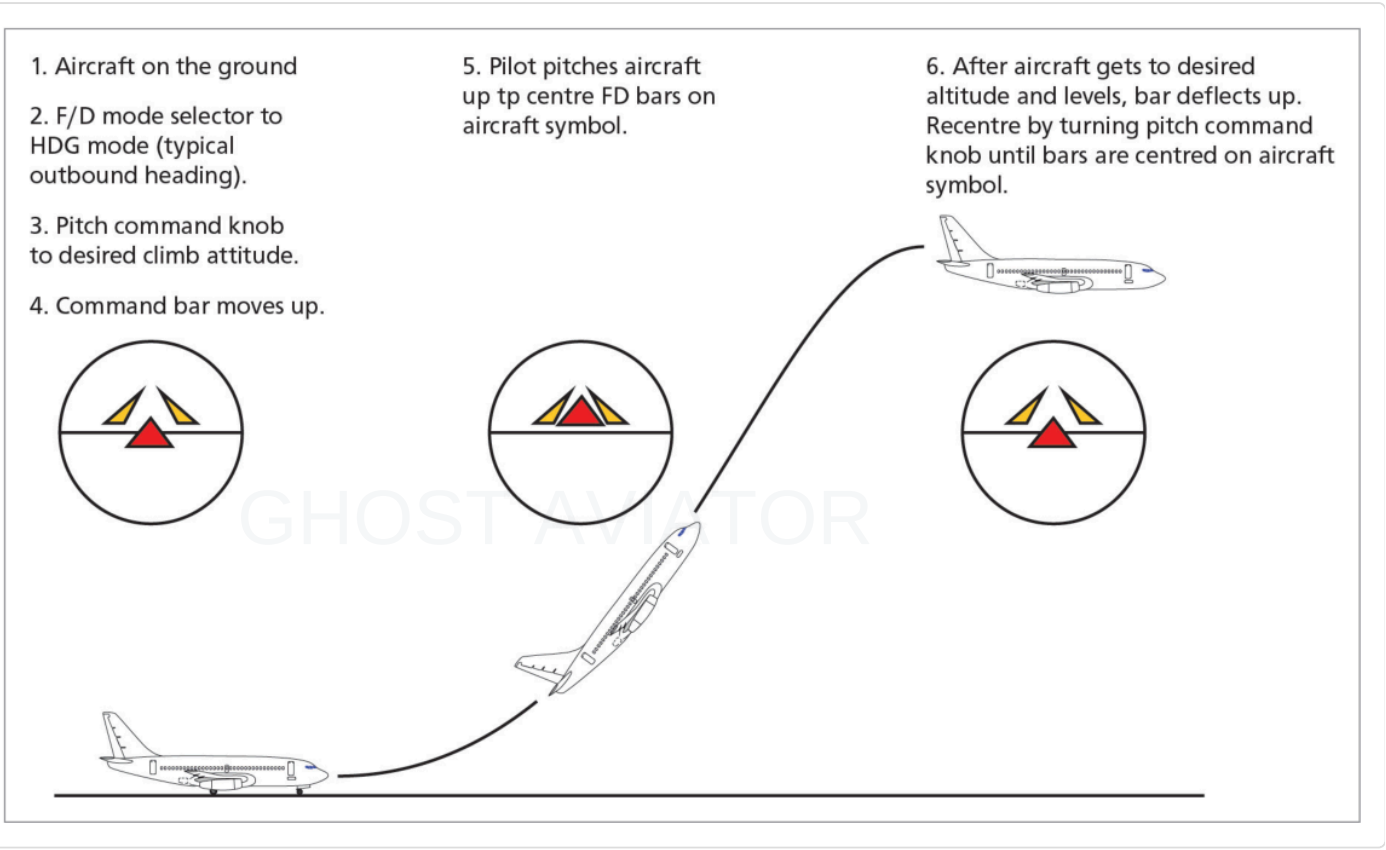
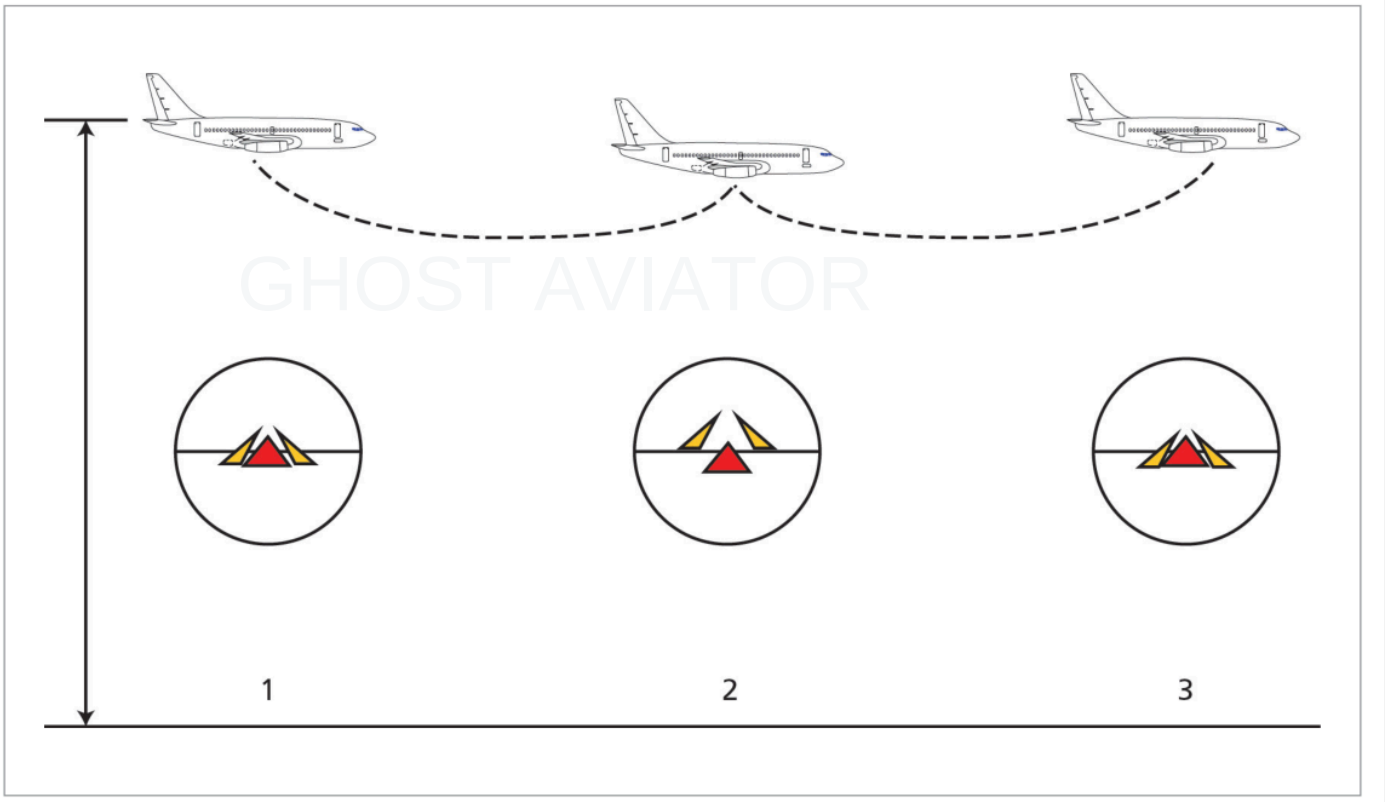


Fig 25.7 — FD modes overview



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Fig 25.8 — Altitude Acquire mode

7. Navigation Modes

> **What this section covers:** Heading mode, LOC/VOR/LNAV tracking, and Flight Director Approach (FDA).

Heading Mode

Pilot selects desired heading via knob on HSI, MCP, or central panel. FDS indicates a fly-to command to bring aircraft onto desired heading.

LOC / VOR (LNAV)

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Allows VOR, Localizer, or INS/IRS/FMS/GPS nav information to be fed and displayed on the FDS. After tuning/identifying the VOR, selecting desired track and VOR/LOC on the mode selector, the FDS gives steering commands to intercept and maintain track.

> **Drift Compensation:** Because the FDS compares actual track with desired track and issues commands to fly the desired track, it effectively compensates for wind drift automatically. No wind velocity input required — it is a natural result of commanding the desired course/track.

Flight Director Approach (FDA)

ILS frequency is tuned and identified; QDM set in the course window; mode selector set to AUTO/APP. Sequence:

```
flowchart TD
  A["APP mode selected\nAnnunciator shows LOC armed"] --> B["Localizer
captured\nAnnunciator shows LOC (green)"]
  B --> C["Approaching GS\nAnnunciator shows GS armed (amber)"]
  C --> D["GS captured\nAnnunciator shows GS (green)\nFDS commands nose-down pitch"]
  D --> E["Continue ILS approach\nGain scheduling applied as\naircraft nears threshold"]
```

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> **Above GS Capture:** If established on LOC past the GS intercept point (above GS), modern systems may automatically adjust to capture from above. Older aircraft: temporarily switch to MAN/GS to force GS capture from above, or to force a fixed LOC intercept angle.

8. FD Take-off Mode

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What this section covers: FD take-off mode commands, engagement criteria, and engine failure procedures (Boeing 737-400 as reference).

Both FD systems should be switched on prior to starting the take-off roll. Take-off mode is engaged by pressing a **TO/GA switch on the throttles**.

Take-off FD Commands

Phase	FD Pitch Command	FD Roll Command
Initial (before 60 kt)	**10° nose-down**	Wings level
After **60 kt** IAS	**15° nose-up**	Wings level
After lift-off	Maintain 15°NP until climb rate achieved, then hold MCP speed **+20 kt**	Wings level

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Engaging FD during take-off roll (FD switches off): Press TO/GA after **80 kt** IAS but before **2000 ft** or **150 seconds** after lift-off — command bars appear automatically.

Terminating TO Mode

- Below **400 ft RA**: both FD switches must be turned OFF
- Above **400 ft RA**: select other FD pitch modes or engage autopilot

Engine Failure During Take-off

Engine Failure Timing	FD Reference Speed
Before V2	V2
After V2 but below V2 + 20 kt	Speed at time of failure
At or above V2 + 20 kt	V2 + 20 kt

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Limitation: Reference speed is never less than V2 for the current flap setting.

9. FD Go-around Mode

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What this section covers: FD go-around engagement criteria, commands, and engine-failure scenarios.

For the 737-400, two criteria must be met:

1. In-flight, below **2000 ft RA**, and NOT in the TO mode
2. TO/GA switch pressed

After GA engagement: command bars appear for both pilots; TO/GA annunciated; MCP IAS/MACH display blanks; airspeed cursors display manoeuvring speed for existing flap.

GA FD Commands

Condition	Pitch Command	Roll Command
2-engine GA	**15° nose-up**	Hold approach ground track at engagement
Single-engine GA	**13° nose-up** initial	Same as 2-engine

Single-Engine GA Target Speed

Engine Failure Timing (relative to GA engagement)	FD Target Speed
Prior to GA engagement	MCP selected approach speed
Within 10 seconds of GA; airspeed within 5 kt of GA engagement speed	Airspeed at GA engagement
After 10 seconds; airspeed more than 5 kt above GA engagement speed	Current airspeed at failure

Minimum GA target speed: Never less than V2 speed based on flap position (except in windshear conditions).

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10. FD Manoeuvre Protection & Gain Scheduling

What this section covers: How the FDC protects against over-stress, and how gain scheduling adapts FDS commands during approach.

FD Manoeuvre Protection

Modern FDC has aircraft performance parameters stored in memory. With inputs from ADC and other systems, it ensures it never commands a manoeuvre that would overstress the aircraft. This is the precursor to fly-by-wire

envelope protection.

Flight Director Gain Scheduling

The FDC varies ("schedules") the gain of pitch and roll demands relative to the task. Most significant during the FDA (Flight Director Approach).

Why it's needed: The ILS glide slope beam diverges from the runway. At different distances, the same 1° of beam error represents very different vertical distances:

Distance	1° GS error = vertical distance
6 NM	**~608 ft**
½ NM	**~54 ft**

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As the aircraft nears the transmitter, the required correction magnitude decreases. Gain scheduling reduces commanded corrections accordingly — from nearly full authority early in the approach to perhaps **½ or ⅓** of original value near the threshold.

Gain Scheduling Methods (in order of sophistication)

Method	Trigger	Notes
Time-based	Fixed time after GS capture (e.g. 45 seconds)	Earliest; no actual position data
Marker beacon	Passing inner/outer markers	Stepped; now in disuse — markers removed from many airfields
Radio Altimeter	Actual RA height	Most accurate; gradual, continuous; no ground signals needed

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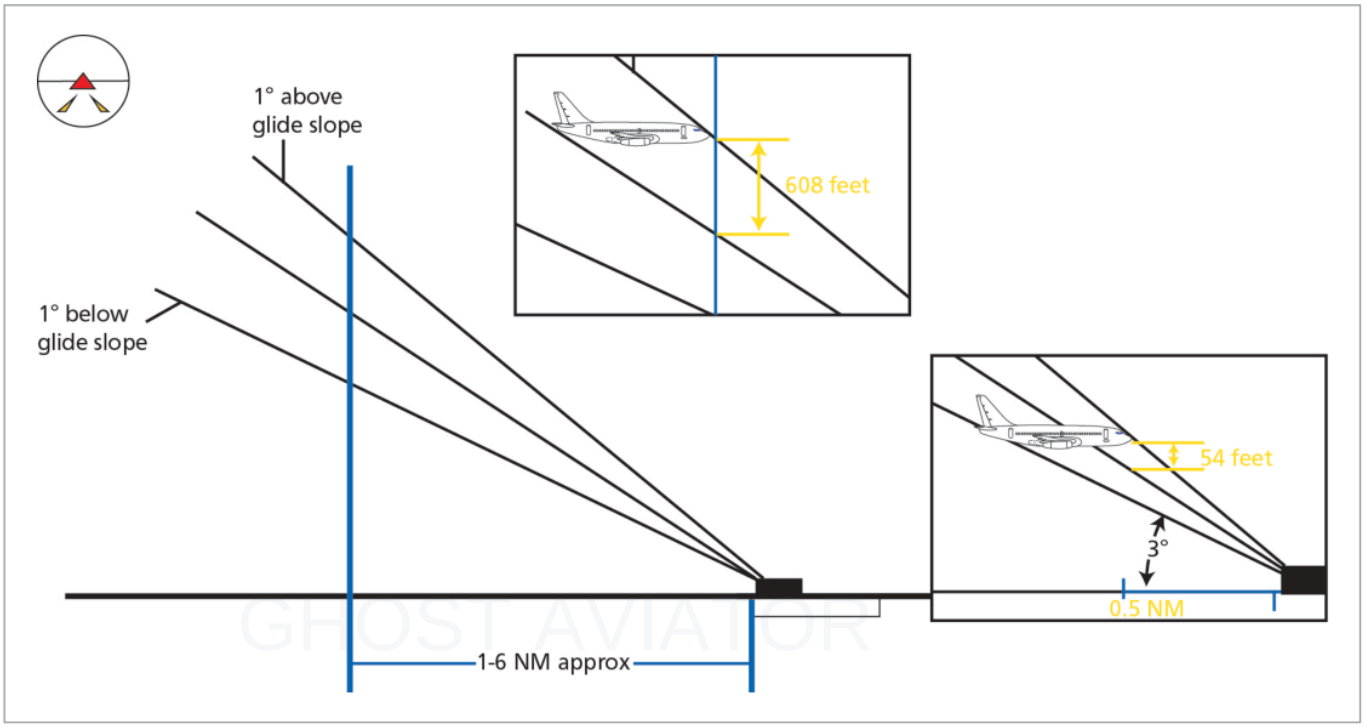


Fig 25.9 — FD gain scheduling during FDA

Exam Tip: Radio altimeter-based gain scheduling is the best method because it is continuous (not stepped), requires no ground infrastructure, and directly correlates with actual aircraft height.

11. Dual FDS & Comparator

What this section covers: Dual FDS on large aircraft — the FD comparator function and redundancy.

Large aircraft with 2 FDS (one per pilot) can use the **FD Comparator** to monitor command bar positions of both systems.

FD Comparator — Command Bars Removed When:

- Pitch difference between the 2 FDS ≥ **1–4°**, OR
- Roll difference between the 2 FDS ≥ **3–9°**

Command bars reappear when difference returns within limits.

Comparator Active Conditions

- Both FD switches ON and neither autopilot engaged
- Only active in TOGA or APP mode below **800 ft RA**

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Comparator Inhibited When:

- On the ground
- Either FD affected by electrical bus transfer
- Failure of either FD sensor or FD itself

Redundancy: If one FDS fails, as long as display equipment is serviceable, both sets of ADI/HSI can be driven from the one serviceable FDC output — often via a simple switch. EFIS aircraft have a spare symbol generator for additional redundancy.

Quick Revision Summary — Chapter 25:

- FDS provides integrated attitude/nav steering commands to pilot and/or autopilot via command bars
- FDS has 2 channels: roll (first/primary) and pitch (second/secondary)
- Raw info = unprocessed deviation; Computed info = rate-of-change processed (anticipates corrections)
- FDC failure: computed info lost, raw info may still be available
- TO/GA engagement: 10° NP initial → 15° NP after 60 kt; TO mode exit below 400 ft RA needs both FD switches OFF
- GA pitch command: 15° NP (2-engine), 13° NP initial (single engine)
- Gain scheduling: reduces FD corrections as aircraft approaches ILS transmitter; RA-based = best method
- FD Comparator: removes bars at 1–4° pitch or 3–9° roll difference; only active in TOGA/APP below 800 ft RA
- FMA: green = engaged (top line), white = armed (second line)

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Practice Questions & Detailed Answers

Instructor-generated questions in DGCA CPL/ATPL examination style.

Q1. The Flight Director System was originally developed as an aid for which phase of flight?

a. Cruise navigation

b. Landing

c. Take-off roll

d. Engine-out climb

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Correct Answer: **(b)** Landing

Explanation: The FDS was originally developed as an aid to the pilot during landing. It provides steering and attitude signals on one instrument to reduce workload during the approach. See Section 1.

Why other options are wrong:

- **(a)** Cruise navigation was a later capability added as FDS technology advanced.
- **(c)** Take-off mode was added later; the original purpose was landing aid.
- **(d)** Engine-out procedures are a specific use case, not the original design purpose.

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Instructor's Note: *The evolution from a landing aid to a full-flight FDS integrated with autopilot is important context for understanding its architecture.*

Q2. What is the key difference between "raw" and "computed" information in an FDS?

a. Raw information comes from the IRS; computed information comes from the FMC

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b. Raw information shows current deviation; computed information uses rate of change to anticipate corrections

c. Raw information is used only for autopilot; computed information is for the pilot only

d. Computed information requires ground station input; raw information does not

Correct Answer: **(b)**

Explanation: Raw information is unprocessed data (e.g., ILS deviation in degrees) — it shows how far off track you are. Computed information is derived from the rate of change of deviation, allowing the FDS to anticipate turns/climbs/descents for smooth interception of the track. If the FDS fails, raw info may still be available. See Section 4.

Why other options are wrong:

- **(a)** Source sensors are not the distinction — both can use ILS/VOR inputs.
- **(c)** Both can be coupled to autopilot or displayed to the pilot.
- **(d)** Computed information is derived onboard from the FDC; no ground station is involved.

Instructor's Note: *On FDC failure, tell your students to use the raw GS/LOC needles — they still work independently of the FDC.*

Q3. On an EFIS FMA, how are engaged modes displayed compared to armed modes?

a. Engaged modes — amber, second line; Armed modes — green, top line

b. Engaged modes — green, top line; Armed modes — white, second line

c. Engaged modes — white, top line; Armed modes — blue, second line

d. Both engaged and armed modes are shown in green; armed modes are boxed

Correct Answer: **(b)**

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Explanation: On an EFIS FMA, engaged modes are shown on the top line in green, while armed modes are on the line below in white. Newly changed information is boxed for emphasis. See Section 3.

Why other options are wrong:

- **(a)** Reverses the green/amber and line positions — engaged is always top line green.
- **(c)** White is used for armed modes, not engaged; blue is not a standard FMA colour.
- **(d)** Armed modes are white, not green; the boxing indicates newly changed modes.

Instructor's Note: *Green = active/working RIGHT NOW. White = standby/waiting to capture. This convention is standard across Boeing and most modern aircraft.*

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Q4. During a Boeing 737-400 take-off with FD engaged, what pitch command is given when IAS reaches 60 knots?

a. 10° nose-up

b. 15° nose-down

c. 15° nose-up

d. 5° nose-up, wings level

Correct Answer: **(c)** 15° nose-up

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Explanation: Initially the FD commands 10° nose-down pitch and wings level. At 60 kt IAS, the FD command changes to 15° nose-up and wings level. See Section 8.

Why other options are wrong:

- **(a)** 10° nose-up is not a standard FD TO command; 10° nose-down is the initial command before 60 kt.
- **(b)** 15° nose-down would put the aircraft on the ground — obviously incorrect.
- **(d)** 5° nose-up is not a standard FD TO command value.

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> ***Instructor's Note:*** *Remember the sequence: 10° ND (initial) → 15° NP (at 60 kt) → then pitch to maintain MCP speed +20 kt after achieving climb rate.*

Q5. Why is Radio Altimeter-based gain scheduling considered superior to time-based or marker beacon-based methods?

a. It is cheaper to install and maintain

b. It provides continuous, gradual scheduling based on actual aircraft height without requiring ground infrastructure

c. It removes the need for an ILS entirely

d. It is required by ICAO for all Category 1 approaches

Correct Answer: **(b)**

Explanation: RA-based gain scheduling is continuous and gradual (vs. stepped), relies on no ground signals (unlike marker beacons, now removed from many airfields), and directly correlates with actual aircraft height above ground — the most operationally relevant parameter. See Section 10.

Why other options are wrong:

- **(a)** Cost is not the cited advantage; the superiority is operational accuracy.
- **(c)** The ILS is still required; gain scheduling just optimizes how the FDS responds to GS deviations.
- **(d)** No specific ICAO requirement for RA-based scheduling is cited; it is a design preference.

> ***Instructor's Note:*** *The fact that marker beacons have been removed from many airfields makes marker-based gain scheduling increasingly obsolete.*

Q6. The FD Comparator removes command bars when the pitch difference between the two FDS is approximately:

a. 5–10°

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b. 10–15°

c. 1–4°

d. ½–1°

Correct Answer: **(c)** 1–4°

Explanation: The FD Comparator monitors command bar positions and removes bars when a difference of approximately 1–4° of pitch and/or 3–9° of roll is sensed between the two FDS. See Section 11.

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Why other options are wrong:

- **(a), (b)** These are larger thresholds than specified — the comparator acts early to alert of divergence.
- **(d)** ½–1° is too small; minor disagreements within this range are tolerated.

Instructor's Note: *The comparator only operates in TOGA or APP mode below 800 ft RA — exactly when precision matters most.*

Master Reference Tables

Key Numerical Values — Chapter 25

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Value	Parameter	Section
60 kt	IAS at which FD TO pitch changes from 10° ND to 15° NP	8
10° NP	FD initial TO pitch command (before 60 kt)	8
15° NP	FD TO pitch command after 60 kt; also 2-engine GA pitch command	8, 9
13° NP	Initial FD single-engine GA pitch command	9
80 kt	Min IAS to engage FD during TO with FD switches off	8
400 ft RA	Threshold above which TO mode can be exited by mode selection	8
2000 ft RA	Max RA for GA engagement (with FD switches off: 150 sec after liftoff)	9
10 sec	Critical time threshold for SE GA target speed determination	9
6 NM	1° GS error ≈ 608 ft vertical	10
½ NM	1° GS error ≈ 54 ft vertical	10
1–4°	FD comparator pitch difference trigger	11
3–9°	FD comparator roll difference trigger	11
800 ft RA	Below this RA, FD comparator becomes active in TOGA/APP	11

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